

Instructions. (15 points) Sections 12.6, 13.1–13.2. Show relevant work for full credit.

(9pts) 1. Consider the surface $z = 4 - x^2 - 4y^2$.

(a) (2 pts) Find the horizontal traces ($z = k$). For what values of k do real traces exist?

Solution:

Set $z = k$: then $k = 4 - x^2 - 4y^2$, or equivalently

$$x^2 + 4y^2 = 4 - k \iff \frac{x^2}{4 - k} + \frac{y^2}{(4 - k)/4} = 1.$$

- $k < 4$: **ellipses** (wider in x than y), shrinking as k increases toward 4.
- $k = 4$: just the **origin** (degenerate ellipse).
- $k > 4$: **empty** (no real trace).

(b) (2 pts) Find the vertical traces in the xz -plane ($y = 0$) and the yz -plane ($x = 0$).

Solution:

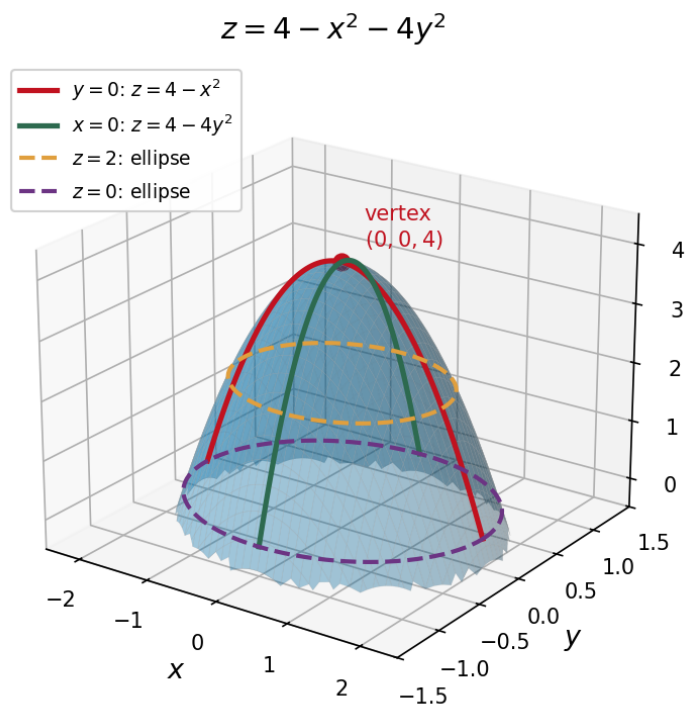
$y = 0$: $z = 4 - x^2$ — a **downward parabola** with vertex at $(0, 4)$.

$x = 0$: $z = 4 - 4y^2$ — a **downward parabola** with vertex at $(0, 4)$, narrower than the first.

(c) (3 pts) Sketch the surface. Label the vertex and at least one trace in each direction.

Solution:

Elliptic paraboloid opening **downward**, with vertex at $(0, 0, 4)$ and axis of symmetry along the z -axis. The xz -trace ($y = 0$) is a downward parabola $z = 4 - x^2$ with x -intercepts at ± 2 . The yz -trace ($x = 0$) is a narrower downward parabola $z = 4 - 4y^2$ with y -intercepts at ± 1 . Horizontal slices are ellipses that widen as z decreases from 4.



- (d) (2 pts) Sketch a contour map (in the xy -plane) showing the level curves for $z = 0$, $z = 2$, and $z = 4$. Label each curve with its z -value.

Solution:

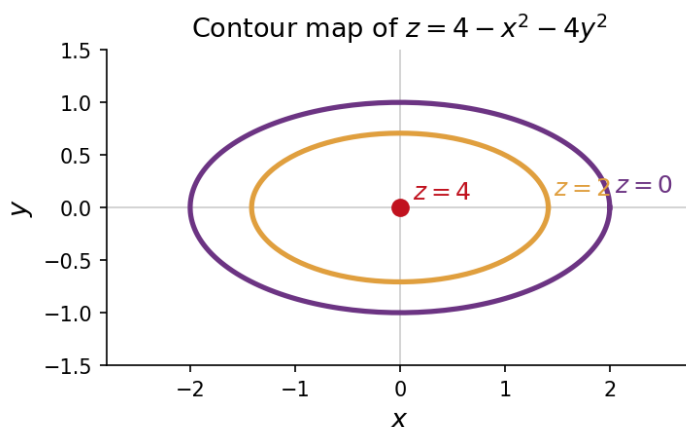
Set $z = k$: $x^2 + 4y^2 = 4 - k$.

$z = 0$: $x^2 + 4y^2 = 4$, i.e. $\frac{x^2}{4} + y^2 = 1$ — ellipse with semi-axes $a = 2$, $b = 1$.

$z = 2$: $x^2 + 4y^2 = 2$, i.e. $\frac{x^2}{2} + \frac{y^2}{1/2} = 1$ — ellipse with semi-axes $a = \sqrt{2} \approx 1.41$, $b = \frac{1}{\sqrt{2}} \approx 0.71$.

$z = 4$: $x^2 + 4y^2 = 0$ — just the **origin** (the vertex, viewed from above).

The contour map shows concentric ellipses shrinking inward toward the origin as z increases. The outermost ellipse is $z = 0$ and the innermost “curve” is the single point $z = 4$.



(6pts) 2. Consider the helix $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 4t \rangle$.

(a) (2 pts) Find the first point (with $t > 0$) where the curve pierces the xz -plane.

Solution:

The xz -plane is where $y = 0$, so $3 \sin t = 0 \Rightarrow t = n\pi$ for integer n . The first positive value is $t = \pi$.

$$\mathbf{r}(\pi) = \langle 3 \cos \pi, 3 \sin \pi, 4\pi \rangle = \boxed{(-3, 0, 4\pi)}.$$

(b) (2 pts) Find the unit tangent vector $\mathbf{T}$ at the point from (a).

Solution:

$$\mathbf{r}'(t) = \langle -3 \sin t, 3 \cos t, 4 \rangle.$$

$$\text{At } t = \pi: \mathbf{r}'(\pi) = \langle 0, -3, 4 \rangle.$$

$$|\mathbf{r}'(\pi)| = \sqrt{0 + 9 + 16} = \sqrt{25} = 5.$$

$$\mathbf{T}(\pi) = \frac{1}{5} \langle 0, -3, 4 \rangle = \left\langle 0, -\frac{3}{5}, \frac{4}{5} \right\rangle.$$

(c) (2 pts) Evaluate $\int_0^{\pi/2} \mathbf{r}(t) dt$.

Solution:

$$\begin{aligned} \int_0^{\pi/2} \langle 3 \cos t, 3 \sin t, 4t \rangle dt &= \left\langle 3 \sin t \Big|_0^{\pi/2}, -3 \cos t \Big|_0^{\pi/2}, 2t^2 \Big|_0^{\pi/2} \right\rangle \\ &= \left\langle 3 - 0, 0 - (-3), \frac{\pi^2}{2} - 0 \right\rangle = \boxed{\left\langle 3, 3, \frac{\pi^2}{2} \right\rangle}. \end{aligned}$$